
A False Belief on a Warehouse Floor

A Relocation One Robot Does Not See, an Observation
Its Beliefs Rule Out, and a Report That Repairs Them

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A crate is moved from one bay to another while the robot that stored it is charging out of sight of both, and the robot that moved it has to reason about where the other will look. The setting follows the false-belief tasks of developmental psychology, the Sally-Anne test in particular, as formalised in dynamic epistemic logic by Bolander. We state it as a planning domain over KD45 models and prove four things about it. A relocation the picker does not observe leaves it believing the crate is where it was, and the mover believing that it believes so. An observation or a truthful announcement that contradicts a belief leaves the believer with no accessible world, so that it believes every formula, and a planner that requires consistent beliefs refuses such an update. A report the picker takes as news of the move repairs its belief consistently and yields common belief of the new location. A picker that allows the move without seeing it is uncertain and not wrong, and finds the crate by looking. The planner reproduces each: with a radio it plans the report, without one it finds no policy, and with a doubting picker it plans a look before the fetch. All three floors were executed in Gazebo through ePlanSys. Without the report the picker drove to the bay it believed the crate was in, the bay the mover's model predicted, and found it empty.

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1 Scope

The false-belief task asks whether a subject can attribute to someone else a belief the subject knows to be false [7]. In the Sally-Anne version [3], Sally puts a marble in a basket and leaves; Anne moves it to a box; the subject is asked where Sally will look for it. The right answer is the basket, where Sally believes it is, and not the box, where the subject knows it is. Bolander [4] formalises the task in dynamic epistemic logic: the move is an action Sally does not observe, the model after it has Sally believing something false, and the question is a formula of the form $B_{Anne}B_{Sally}\varphi$ evaluated on that model.

This report puts the task on the pass-through floor of the companion reports, with two robots. The *picker* stores a crate in bay t_1 and goes to charge at a dock from which neither bay is in sight; the *mover* carries the crate to t_3 ; the picker is then to fetch it. What the report establishes:

- The relocation planted a false belief, and the mover’s model attributed it: after the move the crate is in t_3 , the picker believes it in t_1 , and the mover believes the picker believes so (Proposition 1). The planner and the epistemic state compute this model by product update; it is not written down.
- An observation the picker’s beliefs rule out, such as finding t_1 empty, or a truthful announcement of where the crate is, leaves the picker with no accessible world (Proposition 2). A planner that admits such a state finds a policy that relies on it; one that requires consistent beliefs finds none on the floor without a radio, and the planner returns none.
- A report the picker takes as news of the move repairs its belief and yields common belief of the new location (Proposition 3); with a radio the planner places it before the fetch.
- A picker that allows the move without seeing it has no false belief to repair, and finds the crate by looking (Proposition 4).
- All three floors were executed in Gazebo through ePlanSys, and on the floor without a radio the picker drove to t_1 and found it empty (Section 10).

Not established: belief revision. When an observation contradicts a belief, the logic used here has no answer short of the empty relation; plausibility models [1] give one, and are outside what the planner implements. The crate is moved by the simulator once a robot is in position to carry it: the robots have no gripper.

2 Preliminaries

Definition 1 (Epistemic model). Let $\mathcal{A}$ be a finite set of agents and P a finite set of atoms. A model is $\mathcal{M} = (W, R, V, W_d)$ with W a finite set of worlds, $R : \mathcal{A} \rightarrow 2^{W \times W}$, $V : W \rightarrow 2^P$ and $\emptyset \neq W_d \subseteq W$ the designated worlds. It is S5 when every R_a is an equivalence, and KD45 when every R_a is serial, transitive and Euclidean. A formula holds at $\mathcal{M}$ when it holds at every designated world.

Definition 2 (Belief). $\mathcal{M}, w \models B_a\varphi$ iff $\mathcal{M}, v \models \varphi$ for every v with wR_av . For a group G , $C_G\varphi$ quantifies over the worlds reachable from w by one or more steps of $\bigcup_{a \in G} R_a$. Over S5 models B_a is knowledge and is written K_a ; over KD45 models it is belief, which can be false. The tools write both as $[a]$.

Definition 3 (Consistent belief). Agent a has a *consistent belief* at w if $R_a(w) \neq \emptyset$. If $R_a(w) = \emptyset$, then $\mathcal{M}, w \models B_a\varphi$ for every φ , $\perp$ included: a believes everything. A model in which every world reachable from W_d has a successor for every agent is *consistent*.

Definition 4 (Event model and product update [2]). An event model is $\mathcal{E} = (E, T, Q, pre, post, E_d)$ with events E , observability types T , $Q : T \rightarrow 2^{E \times E}$, preconditions, postconditions and designated

events E_d . Each agent a is given a type τ_a for the whole model, the type whose condition holds at the designated worlds, as plank and Aletheia evaluate it. Then $\mathcal{M} \otimes \mathcal{E} = (W', R', V', W'_d)$ with $W' = \{(w, e) : \mathcal{M}, w \models \text{pre}(e)\}$, $(w, e) R'_a (v, f)$ iff $w R_a v$ and $(e, f) \in Q(\tau_a)$, V' given by the postconditions, and $W'_d = \{(w, e) \in W' : w \in W_d, e \in E_d\}$.

Lemma 1. If R_a and $Q(\tau_a)$ are transitive, so is R'_a ; if both are Euclidean, so is R'_a ; if both are equivalences, so is R'_a . Seriality is not preserved: (w, e) has no a -successor whenever no $v \in R_a(w)$ satisfies the precondition of any f with $(e, f) \in Q(\tau_a)$.

Proof. If $(w, e) R'_a (v, f) R'_a (u, g)$ then $w R_a u$ and $(e, g) \in Q(\tau_a)$ by transitivity of each, and $(u, g) \in W'$, so $(w, e) R'_a (u, g)$. If (w, e) reaches both (v, f) and (u, g) then $v R_a u$ and $(f, g) \in Q(\tau_a)$ by Euclideaness of each, so $(v, f) R'_a (u, g)$. Reflexivity and symmetry carry over in the same way on the worlds that survive. The last claim is the definition of R'_a with an empty set of candidates [6]. $\square$

Aletheia offers three treatments of a product that loses seriality: keep it, as plank does (**Ignore**, the default); delete the worlds that lost it (**Repair**); or refuse the update (**Require**, enabled by **-consistent-beliefs**). The planner of this report runs with **Require**, and ePlanSys's planner plugin sets it by default. The epistemic state that tracks the run applies updates with **Ignore** (Section 11).

3 The domain

Agents p (picker) and m (mover); bays t_1 and t_3 ; atoms $\text{crate}(b)$ for each bay, $\text{at-dock}(p)$, fetched , the static facts radio and doubts , and two phase atoms. Initially every atom is common knowledge, the crate is in t_1 , and the model has one world. The goal is fetched .

The shift. Its first two actions are not chosen by the planner. The picker goes to its dock, a public ontic action; then the mover carries the crate from t_1 to t_3 . Two phase atoms hold that order, and every later action requires both cleared.

Definition 5 (Relocation). $\text{Rel}(m, a, b)$ has events δ (the move) and σ (the mover's slot passing with nothing moved), with $\text{pre}(\delta) = \text{crate}(a) \wedge \neg \text{crate}(b) \wedge \text{due}$, $\text{post}(\delta) : \text{crate}(b), \neg \text{crate}(a), \neg \text{due}$, and $\text{pre}(\sigma) = \text{due}$, $\text{post}(\sigma) : \neg \text{due}$. Designated $\{\delta\}$. Three types: $Q(\text{Fully}) = \{(\delta, \delta), (\sigma, \sigma)\}$, $Q(\text{Doubting}) = \{\delta, \sigma\}^2$, $Q(\text{Oblivious}) = \{(\delta, \sigma), (\sigma, \sigma)\}$. The mover is Fully; the picker is Doubting if doubts holds and Oblivious otherwise.

The event σ carries the phase change because the picker knows the schedule: it knows the mover's slot has passed, and does not know the mover used it. A first version left σ trivial, and the phase atom then stayed set in every world the picker considered possible (Section 11).

Definition 6 (Report). $\text{Rep}(m, p, a, b)$ has events ρ (the report) and μ (the move, as the picker takes it), with $\text{pre}(\rho) = \text{radio} \wedge \text{crate}(b) \wedge B_m \text{crate}(b) \wedge B_m B_p \text{crate}(a)$, trivial postconditions, and $\text{pre}(\mu) = \text{crate}(a)$, $\text{post}(\mu) : \text{crate}(b), \neg \text{crate}(a)$. Designated $\{\rho\}$. Types $Q(\text{Fully}) = \{(\rho, \rho), (\mu, \mu)\}$ for the mover and $Q(\text{Recipient}) = \{(\rho, \mu), (\mu, \mu)\}$ for the picker.

The precondition $B_m B_p \text{crate}(a)$ is the false-belief test as a precondition: the mover reports only what it believes the picker believes otherwise.

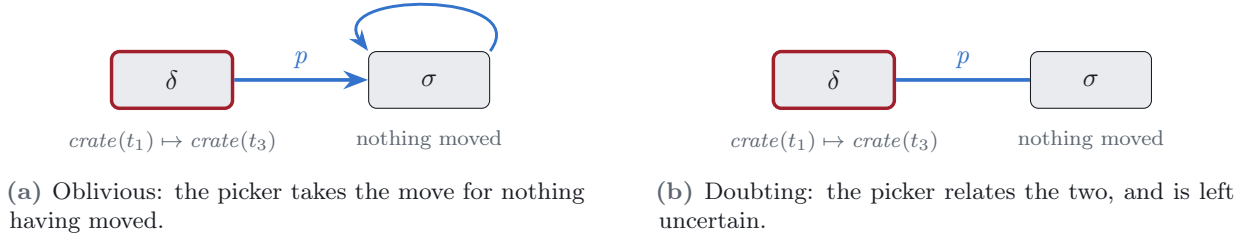


Figure 1: The relocation $Rel(m, t_1, t_3)$ under the picker's two types. Reflexive edges of the mover are omitted; the mover is Fully.

The remaining actions: $look(p, b)$ is semi-private sensing with events $crate(b)$ and $\neg crate(b)$, the picker Fully and the mover Partially; $fetch(p, b)$ is public and ontic with precondition $crate(b) \wedge B_p crate(b)$: the picker drives to the bay only if it believes the crate is there. The three floors are three instances that differ in two lines:

Table 1: The three instances.

instance	radio	doubts
told	true	false
untold	false	false
doubt	false	true

```
(:action-type relocation
:events (?do ?skip)
:observability-types (Fully Doubting Oblivious)
:relations (Fully (:forall (?e - event) (?e ?e))
              Doubting (:forall (?e ?f - event) (?e ?f))
              Oblivious (:forall (?e - event) (?e ?skip)) )
:designated (?do)
:conditions (?do (:non-trivial-postconditions)))

(:action-type report
:events (?told ?revise)
:observability-types (Fully Recipient)
:relations (Fully (:forall (?e - event) (?e ?e))
              Recipient (:and (?told ?revise) (?revise ?revise)) )
:designated (?told)
:conditions (?told (:trivial-postconditions)
              ?revise (:non-trivial-postconditions)))
```

Listing 1: The two action types, in `epddl/beliefs.epddl`.

4 The false belief

Let $\mathcal{M}_0$ be the model after the picker has gone to its dock: one world w , $crate(t_1)$ true, both relations $\{(w, w)\}$.

Proposition 1 (A false belief, and its attribution). Let $\mathcal{M}_1 = \mathcal{M}_0 \otimes Rel(m, t_1, t_3)$ with the picker Oblivious. Then $\mathcal{M}_1$ is KD45 and not S5, and at its designated world

$$crate(t_3) \wedge B_p crate(t_1) \wedge B_m crate(t_3) \wedge B_m B_p crate(t_1) \wedge B_p B_m crate(t_1).$$

Proof. Both events apply at w , so $W_1 = \{(w, \delta), (w, \sigma)\}$ with $crate(t_3)$ at the first and $crate(t_1)$ at the second, and the first designated. The mover is Fully: $R_m = \{((w, \delta), (w, \delta)), ((w, \sigma), (w, \sigma))\}$. The picker is Oblivious: $R_p = \{((w, \delta), (w, \sigma)), ((w, \sigma), (w, \sigma))\}$, serial, transitive and Euclidean, and not reflexive at (w, δ) . At (w, δ) : the picker's only world has $crate(t_1)$; the mover's only world is (w, δ) itself, with $crate(t_3)$, from which the picker's only world has $crate(t_1)$; and the picker's world (w, σ) is the mover's only world from itself, with $crate(t_1)$. $\square$

The last conjunct is the picker believing the mover shares its belief: in the world the picker believes, nothing happened to anyone. The fourth is the answer to the Sally-Anne question. Where will the picker look? Where the mover believes it believes the crate is: t_1 .

5 An observation the belief rules out

Proposition 2 (No consistent update). Let $\mathcal{M}, w \models B_a \neg\psi$, and let $\mathcal{E}$ be an event model in which a is Fully and e an event with $\mathcal{M}, w \models \text{pre}(e)$ and $\text{pre}(e) \models \psi$. Then $R'_a((w, e)) = \emptyset$ in $\mathcal{M} \otimes \mathcal{E}$.

Proof. $R'_a((w, e)) = \{(v, f) : wR_av, (e, f) \in Q(\text{Fully}), \mathcal{M}, v \models \text{pre}(f)\}$. Fully relates e only to e , so $f = e$; and every $v \in R_a(w)$ satisfies $\neg\psi$, hence not $\text{pre}(e)$. $\square$

Corollary 1 (The untold floor). On `untold`, after the shift's two actions, no action leads to a consistent state in which *fetch* can be reached. Hence there is no policy when consistent beliefs are required.

Proof. In $\mathcal{M}_1$ the picker believes $\text{crate}(t_1) \wedge \neg\text{crate}(t_3)$. The report is inapplicable without *radio*. $\text{fetch}(p, t_1)$ needs $\text{crate}(t_1)$, false at the designated world, and $\text{fetch}(p, t_3)$ needs $B_p \text{crate}(t_3)$, false. $\text{look}(p, t_1)$ has, at the designated world, only the event of $\neg\text{crate}(t_1)$, and $\text{look}(p, t_3)$ only that of $\text{crate}(t_3)$; the picker is Fully and believes the negation of each, so by Proposition 2 either look leaves it with no world, and is refused. $\square$

A truthful public announcement of $\text{crate}(t_3)$ falls under the same proposition: every agent is Fully and the picker believes $\neg\text{crate}(t_3)$. That is why the report is not an announcement.

Remark 1. Under `Ignore` the look is applied and the picker believes everything, $B_p \text{crate}(t_3)$ included, which makes $\text{fetch}(p, t_3)$ applicable. Aletheia run without `-consistent-beliefs` returns exactly that policy on `untold`: look into t_1 , then fetch from t_3 , on a belief that holds vacuously. Run with it, it returns none.

6 The report

Proposition 3 (A report that repairs the belief). $\text{Rep}(m, p, t_1, t_3)$ is applicable in $\mathcal{M}_1$, and $\mathcal{M}_2 = \mathcal{M}_1 \otimes \text{Rep}(m, p, t_1, t_3)$ is consistent and satisfies $B_p \text{crate}(t_3) \wedge C_{\{p, m\}} \text{crate}(t_3)$.

Proof. At (w, δ) , $\text{crate}(t_3)$ and $B_m \text{crate}(t_3)$ hold, and $B_m B_p \text{crate}(t_1)$ holds by Proposition 1: ρ applies there, and only there, since $\text{crate}(t_3)$ fails at (w, σ) ; μ applies only at (w, σ) . So $W_2 = \{x, y\}$ with $x = ((w, \delta), \rho)$ designated and $y = ((w, \sigma), \mu)$, and $\text{crate}(t_3)$ holds at both, at y by $\text{post}(\mu)$. The picker relates ρ to μ and μ to μ , and $(w, \delta)R_p(w, \sigma)R_p(w, \sigma)$, so $R_p = \{(x, y), (y, y)\}$; the mover is Fully and $R_m = \{(x, x), (y, y)\}$. Every world has a successor for each agent, both worlds satisfy $\text{crate}(t_3)$, and they are all the worlds reachable from x . $\square$

The picker takes the report as the move it did not see, applied to the world it believes, and its belief is then true. Its relation is still not reflexive: it believes the crate moved just now, and it moved earlier. Nothing the picker believes about the crate is false.

7 Doubt

Proposition 4 (Uncertain, not wrong). Let $\mathcal{M}'_1 = \mathcal{M}_0 \otimes \text{Rel}(m, t_1, t_3)$ with the picker Doubting. Then $\mathcal{M}'_1$ is S5, the picker believes the crate in neither bay, and after $\text{look}(p, t_1)$ with the outcome $\neg\text{crate}(t_1)$ the model is consistent and satisfies $K_p \text{crate}(t_3)$.

Proof. The worlds are those of Proposition 1; the picker's relation is now $\{(w, \delta), (w, \sigma)\}^2$, an equivalence, and the mover's the identity. The two worlds disagree on the crate, so the picker believes neither location. The look's negative event applies at (w, δ) and its positive event at (w, σ) ; the picker is Fully, so the two products are unrelated, and from the designated $((w, \delta), \neg)$ the picker sees only itself, where $\text{crate}(t_3)$ holds. $\square$

8 Planning results

The task is epistemic planning in the sense of Bolander and Andersen [5]. `plank` grounds each instance into 13 atoms and 18 ground actions. Aletheia, with consistent beliefs required, returns:

Table 2: Planner results, with `-consistent-beliefs`.

instance	policy	depth
<code>told</code>	<code>go-dock, relocate, report(m, p, t_1, t_3), fetch(p, t_3)</code>	4
<code>untold</code>	<code>none: space exhausted</code>	3
<code>doubt</code>	<code>go-dock, relocate, look(p, t_1), fetch(p, t_3)</code>	4

Without `-consistent-beliefs` `untold` has the policy of Remark 1. `tools/validate.sh` checks six statements without a simulator: the false belief and its attribution after the shift’s two actions; the told policy and its consistency at every node; no policy on `untold`; the collapse after the look; the vacuous policy without consistent beliefs, and that the trace finds the collapse in it; and the doubt policy, with the picker uncertain and its relation reflexive after the move.

9 Grounding on the floor

The floor is the pass-through floor: a racking block across the hall with three tunnel bays, t_1 , t_2 and t_3 , the storage floor below it and the dispatch floor above. The crate starts in t_1 . The picker works the storage floor and reaches a bay from its south mouth; the mover works the dispatch floor and reaches it from the north. The dock is against the west wall between two rows of racking.

The domain makes three claims about the floor, and `make_floorplan.py` checks each by casting rays over the occupied cells before writing the plan: from the dock neither bay is in sight; from either mouth a robot sees along the bay to the crate’s place and through to the far floor; and every place a robot is sent is reachable over the inflated plan. The performers check the first again at run time: `go_dock` logs which bays are in line of sight once the picker is docked, and `relocate` logs, when it sets the crate down, which bays are in sight of where the picker stands, and refuses if one is.

Looking into a bay is the picker’s laser, read straight along the bay’s axis from its mouth: a crate reads at about three metres, an empty bay through to the dispatch floor. `fetch` reads the same before it drives in, and refuses if the bay is empty, whatever the model says. The crate rides on the robot that carries it, moved through `gazebo_ros_state`.

10 Runs

The three floors were recorded on a 3840×1080 virtual display with RViz on the left and Gazebo on the right, and composed into one film whose captions are the lines the runs wrote. In the tables the model column gives worlds and designated worlds after the update, and the picker’s belief.

10.1 Untold

Table 3: The untold floor, as the run reported it.

action	reported	model
plan	no policy for <i>fetch</i> ed after 0.1 s; space exhausted at depth 3	1 / 1
<i>go-dock</i> (<i>p</i>)	at its dock after 33 s; bays in line of sight: t_1 no, t_3 no	1 / 1
<i>relocate</i> (<i>m</i> , t_1 , t_3)	takes the crate from t_1 after 56 s, sets it down in t_3 after 124 s; in sight of the picker: neither bay	2 / 1, $B_p t_1$
<i>fetch</i> (<i>p</i> , t_1)	belief requirement held, dispatched; at the mouth of t_1 after 34 s the laser read 11.2 m: empty; refused by the performer	refused

The mission then asked the epistemic state four questions and reported the floor as the expected outcome on the answers: the crate is in t_3 ; the picker believes it in t_1 ; the mover believes the picker believes so; the picker believes the mover believes it in t_1 . These are the four conjuncts of Proposition 1, evaluated on the model the executor kept.

10.2 Told

Table 4: The told floor, as the run reported it.

action	reported	model
plan	4 policy items, 1 leaf, 0.1 s	1 / 1
<i>go-dock</i> (<i>p</i>)	at its dock after 35 s; bays in line of sight: t_1 no, t_3 no	1 / 1
<i>relocate</i> (<i>m</i> , t_1 , t_3)	takes the crate from t_1 after 57 s, sets it down in t_3 after 128 s; in sight of the picker: neither bay	2 / 1, $B_p t_1$
<i>report</i> (<i>m</i> , <i>p</i> , t_1 , t_3)	“the crate was moved from t_1 to t_3 ”, delivered and acknowledged	2 / 1, $B_p t_3$
<i>fetch</i> (<i>p</i> , t_3)	at the mouth of t_3 after 71 s the laser read 3.1 m: the crate; set down at the drop 131 s after fetch began	fetch

10.3 Doubt

Table 5: The doubt floor, as the run reported it.

action	reported	model
plan	4 policy items, 1 leaf, 0.1 s	1 / 1
<i>go-dock</i> (<i>p</i>)	at its dock after 36 s; bays in line of sight: t_1 no, t_3 no	1 / 1
<i>relocate</i> (<i>m</i> , t_1 , t_3)	takes the crate from t_1 after 58 s, sets it down in t_3 after 129 s	2 / 1, neither
<i>look</i> (<i>p</i> , t_1)	at the mouth of t_1 after 33 s the laser read 11.2 m: empty, <i>e</i> -empty	2 / 1, $K_p t_3$
<i>fetch</i> (<i>p</i> , t_3)	at the mouth of t_3 after 46 s the laser read 3.1 m: the crate; set down at the drop 108 s after fetch began	fetch

Every node of the three runs exited cleanly when the launch shut down.

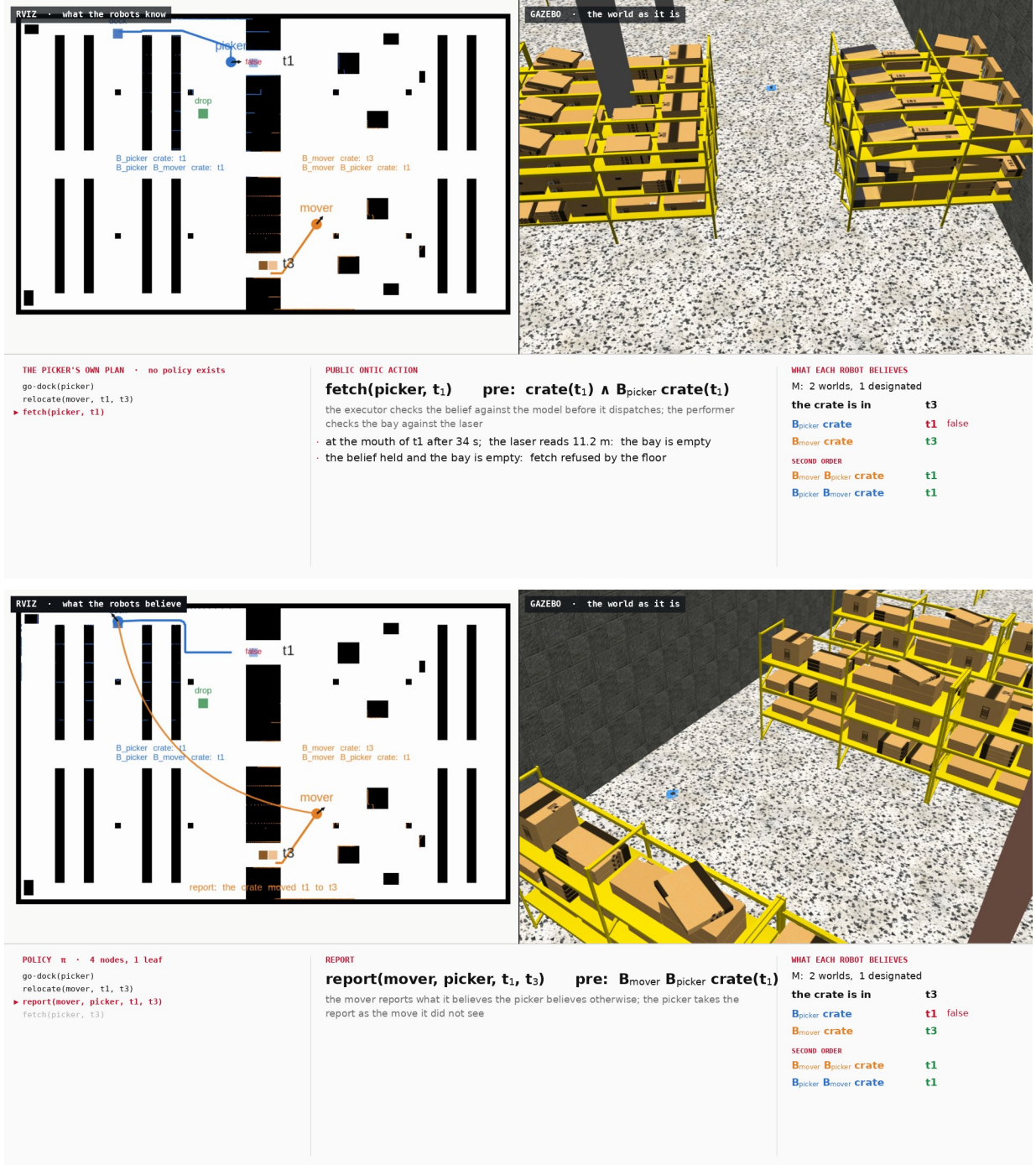


Figure 2: Two frames of the film. Above, the untold floor at the refused fetch: the picker at the mouth of t_1 , its belief of the crate there marked false, and the mover's attribution of it in the second-order rows. Below, the told floor at the report: the arc from the mover to the picker at its dock, a moment before the picker's belief becomes true.

11 Defects

A policy on every floor. The first solve returned the same policy for all three instances: look into t_1 , then fetch from t_3 . Applied by hand, the look left the picker with no world, and the precondition $B_p \text{crate}(t_3)$ of the fetch held vacuously (Remark 1). Aletheia's default keeps such a product, as plank does. With `-consistent-beliefs` the untold instance has no policy, as Corollary 1 says it should not.

A phase the picker could not see pass. With consistent beliefs required, the told instance then had no policy either. The relocation's event σ was trivial, so the phase atom that δ clears stayed set in the world the picker believed, and the fetch, which requires it cleared, had no world to apply to there: the picker's relation emptied at the fetch. The same leak had made the doubt instance's look decisive for the wrong reason, since the look's events also require the phase cleared and so removed the world in which nothing had moved. σ now clears the phase (Definition 5).

The tracked model keeps a collapse. The epistemic state applies updates with **Ignore**. Had the picker looked into t_1 on the untold floor, the tracked model would have left it believing everything, and every belief requirement the executor checked afterwards would have held. The untold run sends the picker to fetch, and the fetch is checked against the floor, which refuses it. A run that observes as well as acts on a false belief would need the epistemic state to refuse the update as the planner does.

12 What the runs establish

That a false belief, and another agent's attribution of it, can be computed by the product update an executor maintains during a run, from an event model in which one robot does not observe a move; and that the attribution predicts where the robot will go. On the untold floor the mover's model placed the picker's belief in t_1 , and the picker drove to t_1 .

That an observation contradicting a belief has no consistent outcome in this logic, and that a planner which refuses inconsistent updates finds no policy where the only way to the goal would pass through one, while a planner which keeps them finds a policy that works only because a robot believes everything.

And that a report taken as news of the move, and not as an announcement of the place, repairs the belief consistently and leaves the location common belief; while a robot that allows a move it does not see is never wrong and needs no report, only a look.

13 Directions for the next iteration

Revision. Plausibility models [1] order the worlds an agent considers by how plausible it finds them, and an observation the most plausible worlds rule out moves belief to the most plausible worlds it allows. With them the untold picker, finding t_1 empty, would come to believe the crate somewhere else, and the untold floor would have a policy: look, then fetch from wherever revision puts the crate.

A mover with a reason. The report is in the policy because the planner needs it. A mover whose own goal is served by the picker succeeding, and whose report has a cost, would put the false-belief test in the agent's decision and not in the planner's search.

The tracked model. The epistemic state could refuse an update that leaves a robot with no world, as the planner does, and report it as a reason to replan.

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Reproduction

```
ros2 launch false_belief_demo false_belief_launch.py floor:=told
ros2 launch false_belief_demo false_belief_launch.py floor:=untold
ros2 launch false_belief_demo false_belief_launch.py floor:=doubt
bash tools/validate.sh # six checks, no simulator
bash tools/record_demo.sh told /tmp/raw_told.mkv /tmp/run_told.log
bash tools/record_demo.sh untold /tmp/raw_untold.mkv /tmp/run_untold.log
bash tools/record_demo.sh doubt /tmp/raw_doubt.mkv /tmp/run_doubt.log
```